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International Mathematics Assessments for Schools

2020 ~ 2021 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2020 ~ 2021 UPPER PRIMARY DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. Which one among the choices has a value greater than $\frac{1}{3}$?

   (A) 0.25 (B) 0.33 (C) 40% (D) $\frac{2}{7}$ (E) 30%

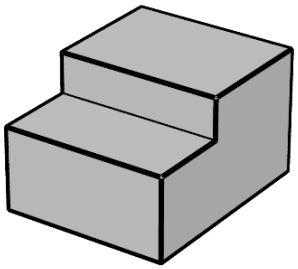
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2. Select one number among the choices and fill it into the box such that the value of $\frac{17}{\square}$ is closest to 2.

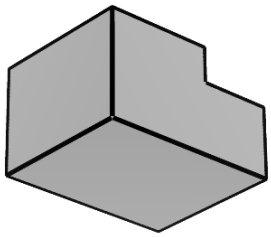
   (A) 6 (B) 9 (C) 11 (D) 13 (E) 15

---

3. The figure below on the right shows the three invisible faces of the three-dimensional solid below on the left. How many edges does this three-dimensional solid have?



View from the right



View from the left

   (A) 15 (B) 16 (C) 17 (D) 18 (E) 19

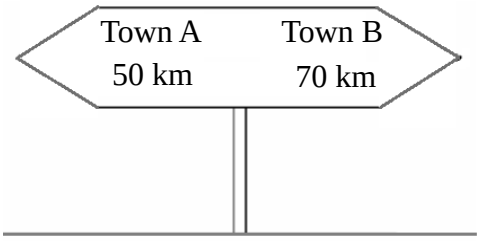
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4. It is known that the product of two positive integers is 120. What is the smallest possible value of the positive difference between these two integers?

   (A) 0 (B) 2 (C) 7 (D) 14 (E) 19

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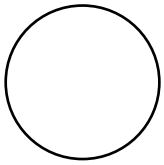
5. Albert drove from Town A to Town B at a constant speed of 80 km per hour. If the street sign shown below was seen during the trip, how many hours does it take Albert to drive from Town A to Town B?

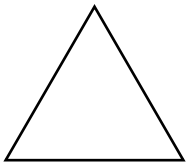


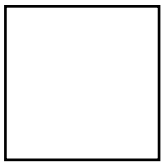
   (A) 2 (B) 1.5 (C) 1 (D) 0.875 (E) 0.625

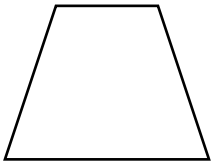
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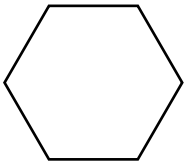
6. Among the figures below, which has the most number of lines of symmetry?

（A） 

（B） 

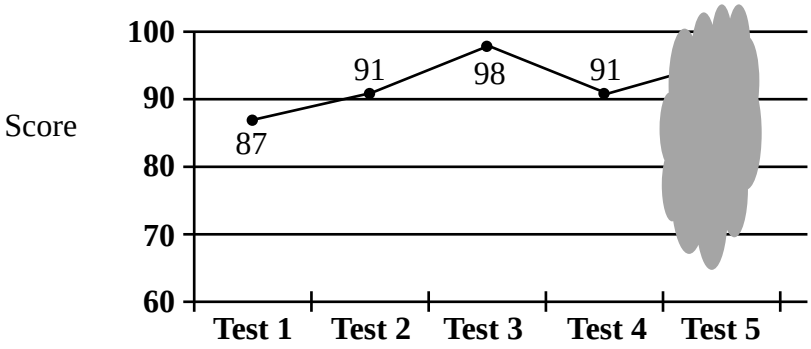
（C） 

（D） 

（E） 

7. The graph below shows the five most recent test scores that Danny got in Mathematics. The score of his fifth test was accidentally stained with ink on the graph and could not be seen. If Danny knows that his average score for these five tests is 92.4, What was his score on the fifth test?

Distribution Chart of Danny's 5 Most Recent Test Scores in Mathematics



（A）93　（B）94　（C）95　（D）96　（E）97

8. Among the shapes listed below, which cannot always be formed by using two identical triangles?

（A）Rhombus　（B）Trapezoid　（C）Square
（D）Rectangle　（E）Parallelogram

9. It is known that the sum of a positive integer and 4 is a multiple of 3. If the sum of the same positive integer and 3 is a multiple of 7, what is the smallest possible value of this positive integer?

（A）4　（B）7　（C）8　（D）11　（E）32

10. A computer is copying a file at a constant speed. If 1 minute and 45 seconds passed and 79% of the file is still uncopied, how long does it take to completely copy the file?

Time elapsed: 1 minute 45 seconds, Percent Until Completion: 79%



（A）8 minutes 20 seconds　（B）7 minutes 09 seconds
（C）5 minutes 45 seconds　（D）3 minutes 04 seconds
（E）2 minutes 13 seconds

Questions 11-20, 4 marks each

11. A box contains some number of red and blue balls. If the number of red balls is two-fifths of the number of blue balls, which of the following numbers can't be the total number of coloured balls in the box?

（A）43　（B）49　（C）63　（D）91　（E）133

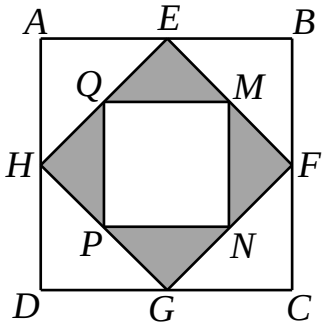
12. Express 2020 as a sum of 290 prime numbers such that the largest prime number used must be as small as possible. What is the largest prime number among these 290 prime numbers?

（A）2　（B）5　（C）7　（D）11　（E）13

13. John can draw a total of 100 sketches in two days. Today, his boss asks him to draw 100 sketches, the only problem is that he will be attending a wedding banquet tomorrow afternoon, so all sketches must be drawn within a day and a half. If he plans to hire a junior artist to work with him together to finish the project in time, how many days at most should the junior artist take to draw 100 sketches?

（A）3　（B）4　（C）5　（D）6　（E）8

14. In the figure below, *ABCD* is a square with an area of 2020 cm^2. Connect the midpoints of each side to get square *EFGH* and then connect the midpoints of each side of square *EFGH* to get square *MNPQ*. What is the area, in cm^2, of the shaded region?



（A）400　（B）404　（C）450　（D）500　（E）505

15. There are three boxes that are painted red, yellow and blue respectively. We are to put 2 identical apples and 2 identical pears into these three boxes such that no two identical fruit can be placed into the same box. If empty boxes are allowed, how many different placement methods can we have?

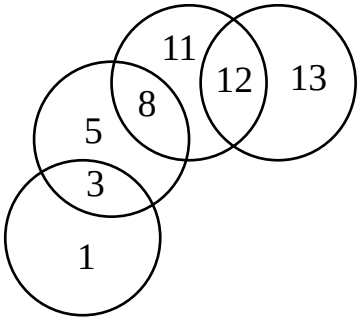
（A）27　（B）18　（C）12　（D）9　（E）6

16. Two bottles of volume 350mL are filled with a mixture of water and alcohol. It is known that the alcohol in the first bottle is 4 times that of water, while the alcohol in the second bottle is 6 times that of water. Now pour both mixtures into a larger empty container. What is the percentage of alcohol in the new mixed solution? (Round off your final answer to one decimal place).

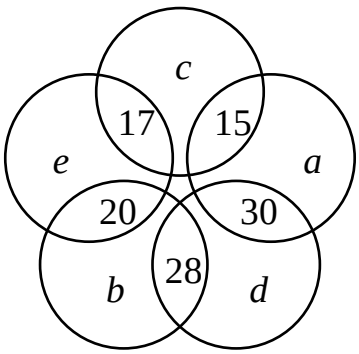


（A）75.0 %　（B）80.0 %　（C）81.7%　（D）82.3%　（E）82.9%

17. Some overlapping circles divide the plane into several regions. Numbers are filled in these areas such that for any pair of overlapping circles, say A and B, the number in the intersection area of circle A and B is the average of the two numbers in the areas solely in A or B. For example, the figure on the right shows $3=\frac{1+5}{2}$, $8=\frac{5+11}{2}$ and $12=\frac{11+13}{2}$.

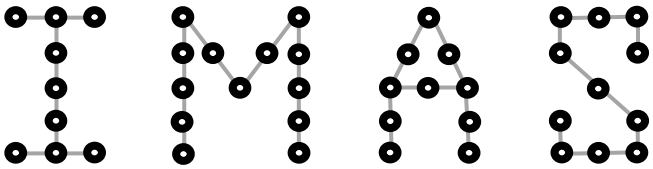


In the figure below, there are five overlapping circles and ten numbers are filled according to the conditions above. What is the sum of the numbers *a* and *b*?



（A）36　（B）40　（C）44　（D）48　（E）52

18. There are 43 pearls with a hole in the center that are to be stringed together using a thread such that the thread will form the letters "I", "M", "A", and "S", as shown in the figure below. If each pearl can only be passed through once by the thread, how many from the four letters cannot be done?



(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

19. The diagram below shows a bicycle model. Its front gear has 24 teeth while its rear gear has 10 teeth. At least how many revolutions does the front gear need to make such that the front and rear gears go back to their initial positions at the same time?





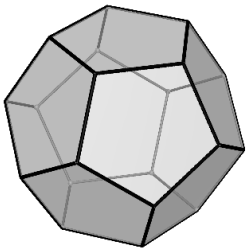
Front Gear



Rear Gear

(A) $\frac{12}{5}$ (B) 5 (C) 10 (D) 12 (E) 20

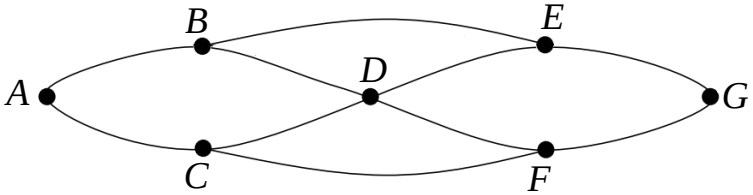
20. Andy used some number of regular pentagonal cardboards of the same size to form a solid which is shown on the right diagram. How many edges does this solid have?



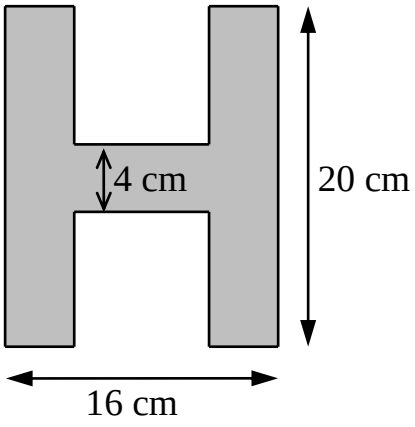
(A) 10 (B) 15 (C) 20 (D) 25 (E) 30

Questions 21-25, 6 marks each

21. The figure below shows the road network between seven cities *A*, *B*, *C*, *D*, *E*, *F* and *G*. How many different routes are there starting from city *A* to city *G*?
(Note: Each section of the network must be a one-way direction from left to right, for example: you cannot go from city *D* then to city *B*)

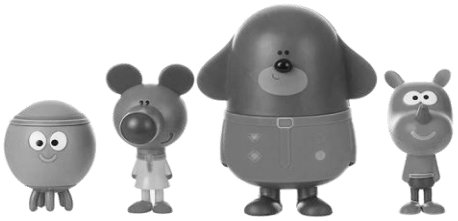


22. The figure below shows one *H*-shaped paper, where each corner is a right angle. What is the perimeter, in cm, of the shaded region?

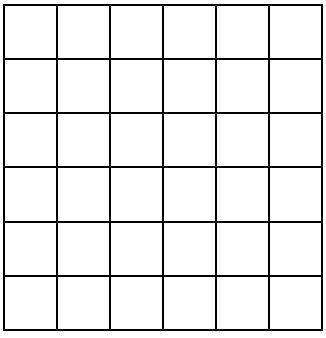


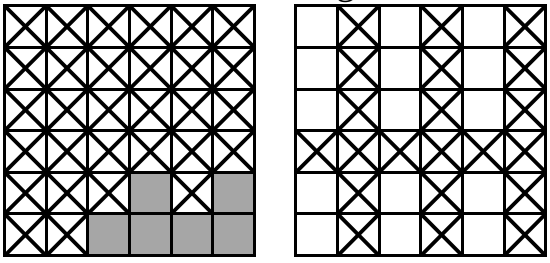
23. Among all positive integers less than 1000, how many even numbers have a remainder of 2 when divided by 3 and a remainder of 1 when divided by 5?

24. Grace wants to buy a total of 10 puppets, which have four kinds namely: Octopus, Mouse, Rhino and Dog puppet using exactly \$700 that she has. The unit prices of these four puppets are \$30, \$50, \$70 and \$110, respectively. If she must buy at least one type of each puppet, how many different purchase combinations are there in total?



25. We need to remove several unit squares from a 6×6 grid such that there does not exist any area that have 6 unit squares which are connected via common sides. What is the minimum number of unit squares that we must remove? For example, in the lower left figure below, the shaded area is an example of an area that have 6 unit squares which are connected together; while the lower right figure shows an example of a grid which does not have any 6 unit squares that are connected together left.





* * *